Lukas Schneider

Charlotte, NC | (874) 989 - 3577 | Lukas.S.055@gmail.com

Creative mathematical and IT professional with 7+ years of experience in research and development of mathematical models and of optimization algorithms with the ability to find interdisciplinary solutions. Excellent interpersonal, organizational, and time management skills with a demonstrated ability to communicate and collaborate with a team.

PROFESSIONAL EXPERIENCE

TNBR Enterprises Inc. — Alpharetta, GA
Owner — February 2021 - March 2024

- Spearheaded search for commercial property for remodeling franchise, oversaw showroom build-out from designing to permitting and inspections, and negotiated and reviewed commercial lease and commercial construction contract.
- Managed sales activities and secured high dollar projects; achieved sales target of $400k in first year.
- Forged partnerships with top industry vendors to provide clients with high-quality products.
- Programmed project planning software based on Python/Django using selenium which saved 30% of project planning and accounting time.

Benjamin Systems GmbH — Darmstadt, Germany
Algorithm Developer, Mathematical Research Scientist — April 2019 – January 2021

- Led the development of mathematics used in a real time navigation software allowing objects to hover efficiently in an electromagnetic field which is used for a wide range of applications, including automated warehouse logistics platforms, airports, manufacturing facilities, and vertical farming facilities.
- Programming in Java, Python, AMPL using and improving modern optimization and pathfinding algorithms, I decreased the cost of logistics by over 50% compared to the industry standard of automated warehouse facilities.
- Successfully applied optimization, statistics, graph-theory and algebra to develop highly efficient algorithms and databases for exponentially growing optimization problems.
- Worked as research mathematician on mathematical theory to build completely new algorithms.
- Presented results to cross-functional teams and the CEO.

INNO3 — Mainz, Germany
Software Developer — October 2019 – July 2020

- Planned and programmed an AI web site, where students study interactively using Python/Django.
- Developed a chatbot for an educational research platform which answers questions by web harvesting.

Chemra GmbH — Darmstadt, Germany
Scientific Consultant — February 2018 – March 2019

- Invented five instruments to distribute fluids and gases evenly in space and on the surface.
- Using completely new ideas and avoiding existing patented technology, my invention saved the company 100% of licensing fees. These instruments can be automatically produced on a large scale and later adjusted in size and fine-tuned for special tasks in chemical production.
- Programmed simulations in MATLAB and Python, Matplotlib using numerical methods and differential equations, and built two prototypes.
- Presented my work directly to the CEO who selected one of my products for further development.
- Researched new methods in chemical production, chromatography and ion exchange systems.

Sebastian Seidl

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Passivhaus Institut — Darmstadt, Germany

Software Developer Intern — July 2017 – December 2017

- Developed software in Python/Django which enables architects and engineers to calculate the energy efficiency and cost of a passive house based on the components they use.
- Created a graphical tool that allows employees to maintain the company's database using psycopg2. Translated SQL queries into Python to be used in different SQL as well as PostgreSQL databases.
- Modularized and simplified code written by different programmers over the years using my knowledge in object-oriented programming and Git.

Self-Employed — Darmstadt, Germany

Teacher for Master Students — 2013 - Present

- Taught abstract algebra, Galois theory, complex analysis, number theory, graph theory, statistics, quantum physics, chemistry, genetics, logic, machine learning and theoretical computer science.

Technical University Darmstadt — Darmstadt, Germany

Teaching Assistant — April 2011 – April 2013

- Teaching Assistant for complex analysis and abstract algebra (group theory).

EDUCATION

Technical University of Darmstadt — Darmstadt, Germany

Masters, Mathematics — 2017

PROJECTS, ACADEMIC EXPERIENCE & CERTIFICATIONS

Master Thesis (Number Theory)

- Research in Automorphic Forms, Eisenstein Series, Epstein and Riemann Zeta-Functions to generalize the "Kronecker Limit Formular" and contribute to the research about the Rieman hypothesis.
- Used Fourier Analysis, Complex Analysis, Abstract Algebra and Lie Groups for these generalizations.

Machine Learning and AI

- Academic experience in natural language processing (NLP), computer science, robotics, quantum physics, and deep learning taking additional courses in Technical University Darmstadt.
- Deep Learning Specialization Course, NLP, RNN: Certificate
- Pre-processed data and developed a predictive model using Jupyter Notebook, TensorFlow and Pandas.

SKILLS

- Python, Pandas, TensorFlow, NumPy, Jupyter Notebook, Excel, Java, MATLAB, Mathematica, R, SQL, LaTeX, Linux, bash, SEO, Git, agile practices
- Measure Theory, Hypothesis testing, Martingales, Bayesian Statistics
- Research and analysis, modeling, project management, presentation, problem-solving.
- English – Fluent, German – Native